

Day 14 Quiz

Geometry Summer Credit | Angle of Elevation, Depression, and Trig Modeling

Name:	Date:	Score:
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Directions

Choose the best answer for multiple-choice items. Draw a model and show a trig equation for short-response items.

1. An angle measured upward from a horizontal line of sight is an angle of:
A elevation B depression C rotation D dilation
2. A person looks from the top of a building down to a car. The angle is measured from a horizontal line at the person's location. It is an angle of:
A elevation B depression C reflection D complement
3. A 40-ft tree casts a shadow. If theta is at the tip of the shadow, which equation is correct?
A $\tan(\theta) = 40/\text{shadow}$ B $\tan(\theta) = \text{shadow}/40$ C $\sin(\theta) = \text{shadow}/40$ D $\cos(\theta) = 40/\text{shadow}$
4. A person with eye level 5 ft finds the vertical triangle height to be 32 ft. The object's total height is:
A 27 ft B 32 ft C 37 ft D 160 ft
5. Why can an angle of depression be transferred to the ground-level angle of elevation?
A They are vertical angles B Horizontal lines are parallel, making alternate interior angles congruent C All acute angles are congruent D They are complementary
6. A student stands 36 ft from a flagpole. The angle of elevation to the top is 44 degrees. Find the height.
7. From the top of an 85-ft building, the angle of depression to a bus is 27 degrees. Find the horizontal distance.

Day 14 Quiz

Continued

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| 8. | A person's eye level is 5.5 ft. The person stands 42 ft from a tree and measures an angle of elevation of 34 degrees. Find the total tree height. |
| 9. | A 50-ft cable reaches the top of a pole and makes a 66-degree angle with the ground. Find the pole height. |
| 10. | Spiral review: A right triangle has opposite side 9 and adjacent side 14. Find the acute reference angle. |
| 11. | Error analysis: A student uses $\tan(30)=d/60$ to find the horizontal distance d from a 60-ft tower. Explain the error and write the correct equation. |